IGCSE 0607 Practice Questions — E1.11

Rates

Total Questions: 21 — **Total Marks:** 114 Marks

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1 A train departs at 08 : 15 and arrives at its destination at 09 : 45.
The train travels a distance of 120 km.
Calculate the average speed of the train.

..... km/h **[2]**

2 A train travels 102 km in 1 hour 12 minutes.
Work out its average speed.

..... km/h **[3]**

3 A car travels 120 km in 1 hour and 30 minutes.
Calculate the average speed of the car in km/h.

..... km/h **[3]**

4 **(a)** A car travels a distance of 108 km in 1 hour 30 minutes.
Calculate its average speed in km/h.

..... km/h **[2]**

(b) Change your answer in part (a) into m/s.

..... m/s **[2]**

5 **(a)** A train travels a distance of 120 km in 1 hour and 30 minutes.
Calculate the average speed of the train.

..... km/h **[2]**

(b) A car uses 8 litres of fuel for every 100 km travelled.
Work out how much fuel the car uses to travel 350 km.

..... litres **[2]**

6 **(a)** A train travels a distance of 180 km in 2 hours and 15 minutes.
Calculate the average speed of the train in km/h.

..... km/h **[2]**

(b) An empty tank is filled with water at a constant rate of 1.5 litres per second.
Work out the time it takes to fill the tank with 450 litres of water.
Give your answer in minutes.

..... minutes **[2]**

7 A cyclist travels 30 km at an average speed of v km/h.
He then travels a further 40 km at an average speed of $(v + 5)$ km/h.
The total time taken for the entire journey is 4 hours.

(a) Write down an equation in terms of v and show that it simplifies to $2v^2 - 25v - 75 = 0$.

[3]

(b) Solve the equation $2v^2 - 25v - 75 = 0$ to find the possible values of v.

$v =$ ..... , or $v =$ ..... **[3]**

(c) Find the time taken for the first 30 km of the journey.

..... hours **[1]**

8 A train travels a distance of 385 km.
The journey takes 4 hours 25 minutes.
Calculate the average speed of the train in km/h.
Give your answer correct to 1 decimal place.

..... *km/h* **[3]**

9 Water flows from a pipe into a rectangular tank at a rate of 45 litres per minute.

(a) Calculate the volume of water, in cubic centimetres (cm^3), that flows into the tank in 1 second.

..... cm^3 **[2]**

(b) The tank has a base measuring 1.5 m by 80 cm. Calculate the rate at which the water level is rising in the tank, giving your answer in cm/s.

..... cm/s **[3]**

10 A car uses fuel at an average rate of 6.4 litres per 100 km.

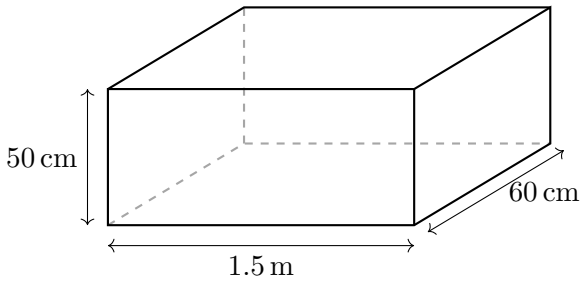
(a) Calculate the amount of fuel used for a journey of 350 km.

..... litres **[2]**

(b) The cost of fuel is 1.35 euros per litre.
The exchange rate is \$1 = 0.85 euros. Calculate the cost of the fuel for this journey in dollars (\$).
Give your answer correct to the nearest cent.

\$ ..... **[3]**

11 A rectangular water tank measures 1.5 m by 60 cm by 50 cm.



(a) Calculate the volume of the tank in litres.

..... litres **[2]**

(b) Water is pumped into the empty tank at a constant rate of 24 litres per minute.
Calculate the time taken to fill the tank completely.
Give your answer in minutes and seconds.

minutes ..... , seconds ..... **[3]**

12 A swimming pool has a volume of 52 m^3.

(a) Water flows into the empty pool from a pipe at a rate of 1.6 litres per second.
Calculate the time taken to completely fill the pool.
Give your answer in hours and minutes, correct to the nearest minute.

hours, minutes **[3]**

(b) The cost of the water is \$2.15 per cubic metre. The exchange rate is \$1 = 0.84 euros.
Calculate the total cost of the water used to fill the pool in euros.
Give your answer correct to 2 decimal places.

euros ..... **[3]**

13 **(a)** A car travels 325 km.
It uses fuel at a rate of 8.4 litres per 100 km.
Fuel costs \$1.35 per litre.
Calculate the total cost of the fuel used for this journey.

\$ ..... **[3]**

(b) On a different journey, the car travels at an average speed of 75 km/h.
The journey takes 4 hours and 24 minutes.
Work out the distance travelled.

..... km **[3]**

14 Water flows from a pipe into a rectangular tank at a rate of 2.5 litres per second.
The tank has a length of 1.8 m, a width of 1.5 m, and a height of 1.2 m.
The tank is initially empty.

(a) Calculate the capacity of the tank in litres.

..... litres **[2]**

(b) Work out how long it takes to fill the tank completely.
Give your answer in minutes.

..... minutes **[2]**

(c) Another pipe empties the full tank at a rate of x litres per minute.
It takes 45 minutes to empty the tank completely.
Calculate the value of x.

$x =$ ..... **[2]**

15 **(a)** Water flows from a pipe into an empty tank at a rate of 0.35 litres per second.
The tank is a cuboid with length 2.5 m, width 1.4 m and height 1.2 m.

Calculate the time taken to fill the tank completely.
Give your answer in hours and minutes.

h ..... , min ..... **[4]**

(b) A mobile phone costs \$850 in the USA and €780 in Germany. The exchange rate is \$1 = €0.84.

Calculate the difference in price between the two countries.
Give your answer in dollars (\$).

\$ ..... **[3]**

16 A car travels from City A to City B, a distance of d km, at an average speed of v km/h.
The car then travels from City B to City C, a distance of $2d$ km, at an average speed of $(v + 15)$ km/h.
The total time taken for the entire journey from City A to City C is T hours.

(a) Show that the total time T can be written as

$$T = \frac{3dv + 15d}{v(v + 15)}$$

[3]

(b) Given that $d = 120$ and $T = 4.5$,

Form an equation in v and show that it simplifies to

$$3v^2 - 195v - 1200 = 0$$

[4]

17 A cyclist travels x km at a speed of 12 km/h. She then travels a further $(x + 5)$ km at a speed of 15 km/h. The total time taken for the whole journey is 2 hours 35 minutes.

(a) Work out the value of x.

$x =$ ..... **[4]**

(b) Calculate her average speed for the whole journey.
Give your answer correct to 1 decimal place.

..... km/h **[3]**

18 An aeroplane flies from City A to City B, a distance of 2400 km.

(a) The aeroplane flies at an average speed of v km/h.
Write down an expression, in terms of v, for the time taken in hours.

..... hours **[1]**

(b) The aeroplane returns from City B to City A.
The average speed on the return journey is $(v + 40)$ km/h.
The total time taken for the two flights is 10 hours and 30 minutes.
Write down an equation in v and show that it simplifies to

$$7v^2 - 2920v - 192000 = 0$$

[4]

(c) Solve the equation $7v^2 - 2920v - 192000 = 0$ to find the average speed of the aeroplane from City A to City B.
Give your answer correct to the nearest integer.

..... km/h **[2]**

19 A cyclist travels from Town A to Town B, a distance of 48 km.
The cyclist travels the first 30 km at an average speed of x km/h. They travel the remaining distance at an average speed of $(x - 4)$ km/h. The total time taken for the journey is 2 hours and 30 minutes.

(a) Show that $5x^2 - 116x + 240 = 0$.

[4]

(b) Solve the equation $5x^2 - 116x + 240 = 0$ to find the possible values of x.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... **[3]**

20 A man runs 15 km at an average speed of x km/h.
He then walks 12 km at an average speed of $(x - 4)$ km/h.
The total time taken for the journey is 3.5 hours.

(a) Show that x satisfies the equation

$$7x^2 - 82x + 120 = 0$$

[3]

(b) **(i)** Solve the equation

$$7x^2 - 82x + 120 = 0$$

$x =$ ..... , or $x =$ ..... **[3]**

(ii) Find the time taken for the man to run the 15 km.

..... hours **[2]**

21 A car travels 150 km at an average speed of x km/h.

(a) Write down an expression, in terms of x, for the time taken in hours.

..... hours **[1]**

(b) The car then travels a further 90 km at an average speed of $(x + 15)$ km/h.
The total time for the whole journey is 3 hours.
Show that $x^2 - 65x - 750 = 0$.

[4]

(c) Solve the equation $x^2 - 65x - 750 = 0$ to find the initial average speed of the car.

$x =$ ..... km/h **[3]**